

Problem

Find all pairs of integers (x, y) satisfying

$$x^3 + y^3 = (x + y)^2.$$

Source: Titu Andreescu

§1 Separating the Trivial Family

Using the factorization

$$x^3 + y^3 = (x + y)(x^2 - xy + y^2),$$

the given equation becomes

$$(x + y)(x^2 - xy + y^2) = (x + y)^2.$$

If

$$x + y = 0,$$

then the equation is automatically satisfied. Hence

$$(x, y) = (n, -n), \quad n \in \mathbb{Z},$$

is an infinite family of integer solutions.

It remains to investigate the case

$$x + y \neq 0.$$

Dividing both sides by $x + y$ yields

$$x^2 - xy + y^2 = x + y.$$

§2 Viewing the Equation as a Quadratic

Treating the equation as a quadratic in x , we obtain

$$x^2 - (y + 1)x + (y^2 - y) = 0.$$

For an integer solution x to exist, the discriminant must be a nonnegative perfect square.

The discriminant is

$$\Delta = (y+1)^2 - 4(y^2 - y) = -3y^2 + 6y + 1.$$

Thus

$$-3y^2 + 6y + 1 \geq 0.$$

Rearranging,

$$3y^2 - 6y - 1 \leq 0.$$

The roots of the quadratic polynomial are

$$y = \frac{6 \pm \sqrt{48}}{6} = 1 \pm \frac{2\sqrt{3}}{3}.$$

Therefore

$$1 - \frac{2\sqrt{3}}{3} \leq y \leq 1 + \frac{2\sqrt{3}}{3}.$$

Since y is an integer, the only possibilities are

$$y \in \{0, 1, 2\}.$$

§3 Checking the Remaining Cases

Case 1: $y = 0$

The equation becomes

$$x^2 - x = 0,$$

so

$$x = 0 \quad \text{or} \quad x = 1.$$

Since $(0, 0)$ already belongs to the family $(n, -n)$, this case contributes

$$(1, 0).$$

Case 2: $y = 1$

The equation becomes

$$x^2 - 2x = 0,$$

so

$$x = 0 \quad \text{or} \quad x = 2.$$

Hence we obtain

$$(0, 1), \quad (2, 1).$$

Case 3: $y = 2$

The equation becomes

$$x^2 - 3x + 2 = 0,$$

which factors as

$$(x - 1)(x - 2) = 0.$$

Thus

$$x = 1 \quad \text{or} \quad x = 2,$$

yielding

$$(1, 2), \quad (2, 2).$$

§4 The Complete Set of Solutions

Combining the infinite family with the exceptional solutions, all integer solutions are

$$\boxed{(x, y) = (n, -n) \ (n \in \mathbb{Z}), \quad (1, 0), \quad (0, 1), \quad (1, 2), \quad (2, 1), \quad (2, 2).}$$